

ISAIAH ALFRED MARTINEZ

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EDUCATION

University of Nebraska - Lincoln

Ph.D. in Mathematics

Advisor: Mark Walker

Lincoln, NE

Aug. 2024 – Present

California State University, Fresno

M.S. in Mathematics

Thesis: Khovanov homology for singular links and tangles

Advisor: Carmen Caprau

Fresno, CA

Aug. 2022 – May 2024

University of California, Santa Barbara

B.S. in Mathematics

Santa Barbara, CA

Sept. 2018 – Mar. 2022

RESEARCH

My current research interests lie in the application of homological methods to commutative algebra. Previously, my research interests were more topological in nature.

PUBLICATIONS

TRI-PLANE DIAGRAMS FOR SIMPLE SURFACES IN S^4

(joint w/ many authors) *Journal of Knot Theory and its Ramifications*, Vol. 32, No. 6

Jun. 2023

SELECTED TALKS

For a comprehensive list along with selected slides/notes/posters, see [my website](#).

Transfer of Koszulness

RTG Seminar on Koszul Algebras

Apr. 2026

Lincoln, NE

Čech vs De Rham cohomology

Cohomology Club @ UNL

Dec. 2025

Lincoln, NE

The tensor product of Hilbert spaces is (almost) a Hilbert space

(joint w/ Audrey Baumheckel) *Operator Theory Reading Seminar (OTRS) @ UNL*

Nov. 2025

Lincoln, NE

Injective resolutions and Bass numbers

Commutative Algebra Reading Seminar (CARS) @ UNL

Sept. 2025

Lincoln, NE

CW and cellular cohomology

(joint w/ Anna Brosowsky) *Cohomology Club @ UNL*

Aug. 2025

Lincoln, NE

An introduction to Khovanov homology

Groups-Semigroups-Topology (GST) Seminar @ UNL

Nov. 2024

Lincoln, NE

Khovanov homology for singular links and tangles

BAMM!Fest 2024

May 2024

San Francisco, CA

An extension of Khovanov homology to singular links

JMM 2024

Jan. 2024

San Francisco, CA

A crash course in Khovanov homology

Department of Mathematics Day, Fresno State

Nov. 2023

Fresno, CA

Tri-plane diagrams and surface knot invariants

44th Annual Central California Research Symposium

Apr. 2023

Fresno, CA

Tri-plane diagrams for simple surface knots

(joint w/ Wolfgang Allred, Zack Dooley, & Alec Goldman) *JMM 2023*

Jan. 2023

Boston, MA

TEACHING EXPERIENCE

University of Nebraska - Lincoln

Aug. 2024 – Present

Instructor of Record

Responsibilities: Full responsibility for course activities including providing weekly lectures and facilitating active learning. Responsible for grading and submitting student grades to the university at the end of each semester. Responsible for holding weekly office hours.

- Math 100A: Intermediate Algebra — Fall 2025

Teaching Assistant

Responsibilities: Responsible for facilitating active learning activities which reinforce the concepts students have seen in lecture in the previous weeks. Responsible for grading student work often including exams and holding weekly office hours.

- Math 107: Calculus II — Fall 2024 (2 sections), Spring 2025 (2 sections), Summer 2025
- Math 208: Calculus III — Spring 2026 (2 sections)

Grader

Responsibilities: Responsibilities vary depending on course needs each semester. Usually this includes grading some subset of homework assignments, quizzes, and/or exams.

- Math 104: Applied Calculus — Spring 2026

California State University, Fresno

Teaching Associate

Aug. 2022 – May 2024

Responsibilities: Responsible for teaching three “lab” courses each semester through a combination of lecture and active learning activities. Responsible for evaluating student assignments and maintaining/submitting student records (grades). Responsible for holding office hours every week.

- Math 3L: College Algebra — Fall 2023
- Math 10AL: Structure & Concepts in Mathematics I — Spring 2024
- Math 11L: Elementary Statistics — Fall 2022, Spring 2023, Fall 2023 (2 classes)
- Math 45L: What is Math? — Fall 2022 (2 classes), Spring 2023 (2 classes), Spring 2024 (2 classes)

University of California, Santa Barbara

Course Reader (Grader)

Mar. 2021 – Dec. 2021

Responsibilities: Provided thoughtful and constructive feedback to homework and assigned grades to every homework assignment. Responsible for submitting grades to “Gauchospace.”

- Math 8: Transition to Higher Mathematics — Spring 2021, Fall 2021

MENTORING

UNL First Year First Generation REU Mentor

Summer 2026

- Served as one of the primary mentors for four incoming UNL undergraduates.
- Organized weekly panel participants to give advice and share first-gen experiences regarding university life.
- Helped organize professional development workshops including $\text{\LaTeX}$ /beamer, coding, and CV writing.
- Helped introduce students to basic set and graph theory so that they could work on a graph theoretical research question that has commutative algebra implications.

UNL Directed Reading Program Mentor

- Fall 2025: Mentee — Eric Woods
- Spring 2025: Mentee — Noah Crawford

UCSB Math Major Peer Mentor with Mentor Collective

Nov. 2021 - June 2022

- Certified trained mentor with Mentor Collective.
- Mentored four undergraduate STEM students in the 2021-2022 academic year, whose concerns ranged from academic performance to seeking employment upon graduation.

ORGANIZATIONAL AFFILIATIONS

American Mathematical Society | *Graduate Student Member*

Sept. 2022 – Present

Math Alliance | *F-GAP Scholar*

Apr. 2022 – May 2024

SERVICE

Nebraska Conference for Undergraduate Wisdom in Mathematics (NCUWM) 2026

Jan 2026

- Gave campus tour to visiting undergraduate participants.

UNL Math Day

Nov. 2025

- Proctored Quiz Bowl rounds.

Nebraska Conference for Undergraduate Women in Mathematics (NCUWM) 2025

Jan. 2025

- Helped organize folders with conference information and welcome bags handed out to participants.

Math Field Day 2023 (Fresno State)

Apr. 2023

- Math field day is an outreach program held annually at Fresno State that attempts to foster mathematical curiosity in local middle school and high school students.
- Assisted with the proctoring of a mathematics contest for high school students.

Sonia Kovalevsky Day 2023 & 2024

Mar. 2023/2024

- Sonia Kovalevsky Day is an annual outreach event held at Fresno State that aims to inspire the next generation of female mathematicians, scientists, engineers, and innovators.
- Assisted with an assortment of mathematics activities and games in an effort to showcase the fun and creative ways one can think about mathematics.

CONFERENCES ATTENDED

CA+ 2026	<i>Mar. 2026</i>
KUMUNU 2025	<i>Oct. 2025</i>
Summer CAMP (Commutative Algebra Market Preparation) @ UNL	<i>Aug. 2025</i>
RayFest (conference celebrating the work and retirement of Ray Heitmann)	<i>Apr. 2025</i>
Commutative algebra in mixed characteristic @ UNL	<i>Mar. 2025</i>
Joint Mathematics Meeting 2024	<i>Jan. 2024</i>
CSU Mathematical Sciences Conference 2023	<i>Nov. 2023</i>
Field of Dreams 2023	<i>Nov. 2023</i>
AMS Spring Western Sectional Meeting 2023	<i>May 2023</i>
44th Annual Central California Research Symposium	<i>Apr. 2023</i>
Joint Mathematics Meeting 2023	<i>Jan. 2023</i>
Field of Dreams 2022	<i>Nov. 2022</i>
Math For All 2021 (hosted virtually by Tulane University)	<i>Mar. 2021</i>
The California Forum for Diversity in Graduate Education	<i>Oct. 2020</i>
Pacific Math Alliance Conference 2019	<i>Oct. 2019</i>

SCHOLARSHIPS, HONORS, AND GRANTS

<u>Fresno State College of Science and Mathematics Outstanding Thesis</u>	<i>2024</i>
<u>Fresno State Department of Mathematics Outstanding Thesis</u>	<i>2024</i>
<u>Graduate Student Ambassador for the Mathematics M.S. program at Fresno State</u>	<i>2022-2023</i>
<u>Fresno State Division of Research and Graduate Studies Graduate Student Travel Grant</u>	<i>2022</i>
<u>Bolstering the Advancement of Masters in Mathematics (BAMM!) Scholarship</u>	<i>2022–2024</i>
Edison GRE Scholarship	<i>2020</i>
UCSB General Scholarship	<i>2018 – 2021</i>
BLOSS MHS Alumni Scholarship	<i>2018,2020,2021</i>

SOFTWARE PROFICIENCY

$\text{\LaTeX}$, Python, Macaulay2, Github